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E-2 OF 3 JANUARY 1939
E-5 OF MARCH, 1939

MILITARY STANDARD

QUALIFICATION TESTS FOR WELDERS

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DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON, D.C.

QUALIFICATION TESTS FOR WELDERS

MIL-STD-248(SHIPS)

1. This standard has been approved by the Bureau of Ships, and is published to establish requirements and tests necessary to qualify welders for production of Military weldments.

2. Recommended corrections, additions, or deletions should be addressed to Chief of the Bureau of Ships, Navy Department, Washington 25, D.C.

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Table II - Grouping of electrodes and welding rods

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MILITARY STANDARD

QUALIFICATION TESTS FOR WELDERS

1. SCOPE

1.1 This standard covers the requirements and tests necessary to qualify welders for production of Military weldments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this standard:

SPECIFICATIONS

MILITARY

MIL-R-908 - Rods, Welding, Steel and Cast Iron (for Gas Welding).
MIL-E-986 - Electrodes (Mineral-Covered, Low Hydrogen), Welding, Medium and Alloy Steels (for Shipbuilding Application).
MIL-E-15599 - Electrodes, Welding, Medium Steel.
MIL-E-15716 - Electrodes (Covered), Welding, Molybdenum Alloy Steel (for Shipbuilding Applications).
MIL-E-16589 - Electrodes, Welding (Chromium Molybdenum and Chromium Alloy Steels, Low-Hydrogen Type).

STANDARDS

MILITARY

JAN-STD-19 - Welding Symbols.
MIL-STD-20 - Welding Terms and Definitions.

PUBLICATIONS

MILITARY

NAVSHIPS 250-692-2 - X-Ray Standards for Production and Repair Welds

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-

3.1.1 Qualification provisions.- Before being permitted to perform any production welding, each welder, (except where qualification is specifically waived (see 3.1.5)), shall demonstrate his ability to produce satisfactorily sound welded joints with the welding process to be used in production. This shall consist of welding the applicable test joints shown in table I to the soundness requirements specified hereinafter for each type of test. (Welder qualification tests are not required for operators of machine-welding equipment.)

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3.1.2 Processes.- The welding processes requiring qualification of welders shall be as follows:

(a) Manual, shielded metal arc process.
(b) Semi-automatic, inert-gas (consumable electrode) metal arc process.
(c) Semi-automatic submerged arc process.
(d) Manual, inert-gas (tungsten electrode) metal arc process.
(e) Manual carbon arc process.
(f) Manual gas welding process.
(g) Manual torch brazing process.

3.1.3 Extent of process qualifications.- Welders qualified for a given process with a specific welding rod or electrode shall be considered qualified for welding by that process with any other rod or electrode of similar type (see 3.4).

3.1.4 Relation of tests to production procedures.- The qualification tests required are devised to demonstrate a welder's ability to produce sound welds and may or may not conform in detail to production procedures. Production procedures shall be as specified by the bureau or agency concerned.

3.1.5 Provisions for waiver of qualification tests.- Qualification tests may be waived under the following conditions:

(a) Qualification tests of welders working on assemblies, the possible failure of which is remote and would not result in danger to the vessel, plant or structure, may be omitted provided this omission is authorized by the bureau or agency concerned. However, welders who are to perform such work shall demonstrate to the satisfaction of the Government inspector their ability to do the proposed work in a satisfactory manner, using the process, equipment and materials intended for use in production.
(b) Qualification tests of metal arc welders working on assemblies may be omitted, where the initial production welds are radiographically inspected, and found satisfactory to the standards in 3.3.2.2 provided this omission is authorized by the bureau or agency concerned. The welder shall be considered qualified for the process, equipment and materials and in the position of the production welding.
(c) Qualification tests of welders may be omitted for welders who weld satisfactory procedure qualification test assemblies as required by the bureau or agency concerned. The welder shall be considered qualified for the process, equipment and materials and in the position for which the procedure is qualified.
(d) Welders who have been qualified by the U.S. Coast Guard, or American Bureau of Shipping shall be considered qualified within the limits covered by the qualification tests passed.

3.1.6 Responsibility of fabricator in test administration.- The fabricator shall direct and supervise the testing of welders and shall bear the expense of conducting the tests. When welders are to be qualified, the fabricator shall notify the Government inspector and arrange a suitable time and place for conducting the tests so the Government inspector may be present.

3.1.7 Requalification requirements.- A welder shall be requalified whenever one or more of the changes listed below are made to be used in production:

(a) When the filler metal is changed to a filler metal having a different M number in table II or to a filler metal not included, except that qualification under M2 shall qualify a welder for M1.
(b) When the welding position is changed to positions other than those already qualified.
(c) When the backing strip in arc-welding single welded butt joints is omitted.
(d) When a backing strip in gas welding is added.
(e) When there is a change of welding processes.

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3.1.8 Duration of qualification. - The welder's qualification from tests herein specified shall remain in effect indefinitely unless:

(a) The welder does not use the process for which he was qualified for a period of three months or more.
(b) The Government inspector believes there is reasonable doubt of the welder's ability. In such cases the welder shall not perform production welding until he passes the applicable qualification tests.

3.1.9 Retests. - If any of the welder's qualification test welds fails to meet the requirements specified hereinafter, he may be allowed a retest under the following conditions:

(a) An immediate retest may be made which shall consist of two test welds of each type he failed, all of which shall meet all the applicable requirements.
(b) A retest may be made provided the welder has had further training or practice. In this case the retest shall be complete.

3.1.10 Test records. - Qualification test records of each welder, together with identification data shall be kept by the fabricator on cards similar to figure 1. Upon request, duplicate records shall be furnished to the Government inspector or any authorized participating agency.

3.1.11 Transferral of qualification. - The qualification of a welder shall not be transferred from one company's organization to another company's organization without the specific approval of the bureau or agency concerned.

3.2 Qualification tests. - The welder qualification tests shall consist of welding the applicable test assemblies shown in table I.

Table I - Welder qualification tests and applicable conditions.

Construction and material size	Position in which welding shall be done Note 1		
	Flat, vertical, horizontal and overhead-grooves and fillets	Flat and vertical-grooves and fillets horizontal-fillets	Flat-grooves and fillets horizontal fillets
On material of limited thickness (3/4 inch or less) Notes 2, 3, and 4	Test No. 1 in vertical and overhead positions (fig. 3)	Test No. 1 in vertical position (fig. 3)	Test No. 1 in flat position (fig. 3)
On material of unlimited thickness Note 5	Test No. 2 in vertical and overhead positions (fig. 4)	Test No. 2 in vertical position (fig. 4)	Test No. 2 in flat position (fig. 4)
On pipe joints 3/4 inch or less thick Notes 2, 6, 7, and 9	Test No. 3 in horizontal and vertical fixed positions (fig. 5)		Test No. 3 in horizontal rolled position (fig. 5)
On pipe joints over 3/4 inch thick Notes 8 and 9	Test No. 4 in horizontal and vertical fixed positions (fig. 5)		Test No. 4 in horizontal rolled position (fig. 5)
For welders doing only tack or fillet welding	Test No. 5 in vertical and overhead fillet positions (fig. 7)	Test No. 5 in vertical and horizontal fillet positions (fig. 7)	Test No. 5 in horizontal fillet position (fig. 7)

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Table I - Welder qualification tests and applicable conditions (cont'd.).

Construction and material size	Position in which welding shall be done Note 1		
	Flat, vertical, horizontal and overhead-grooves and fillets	Flat and vertical-grooves and fillets horizontal-fillets	Flat-grooves and fillets horizontal fillets
For torch brazed piping Note 9	Test No. 6 in horizontal and vertical fixed positions (fig. 8)		
For torch brazing other than piping	Test No. 7 in vertical and overhead fixed positions (fig. 9)		

Note 1. - In special cases when welding is to be done in positions other than those shown in figure 2 the welder qualification may be made in the position in which welding will be done with the limitation that the qualification shall be valid only for the actual position tested. An angular tolerance of plus or minus 15 degrees shall apply.

Note 2. - Where the maximum thickness of material on which a welder may work throughout the period governed by a test is indeterminate, the Government inspector may, if desired, require the welder to qualify under unlimited thickness requirements.

Note 3. - Where the maximum thickness of material to be welded throughout the period governed by the test is less than 3/8 inch, the fabricator may, with the permission of the Government inspector, use the following modification of test number 1: The thickness of material shall be the maximum to be welded in production but shall not be less than 3/16 inch. The root opening shall not be more than one nominal diameter of the filler metal to be used. The backing strap thickness shall be not less than 5/32 inch.

Note 4. - Welders qualified under the requirements of test number 1 in the flat position shall be considered qualified to weld piping 3/4 inch or less thick in the horizontal rolled position and longitudinal pipe joints 3/4 inch or less thick in the flat position. Welders qualified by test number 1 in the vertical position shall be considered qualified to weld piping 3/4 inch or less in thickness in the horizontal fixed position from the horizontal axis up.

Note 5. - Welders qualified under the requirements of test number 2 in the flat position shall be considered qualified to weld piping over 3/4 inch thick in the horizontal rolled position and longitudinal pipe joints over 3/4 inch thick in the flat position. Welders qualified by test number 2 in the vertical position shall be considered qualified to weld piping over 3/4 inch thick in the horizontal fixed position from the horizontal axis up.

Note 6. - Where the maximum wall thickness of pipe to be welded throughout the period covered by the test is less than 3/8 inch, the fabricator may with the permission of the Government inspector use the modified test number 3A where the wall thickness of the pipe shall be the maximum to be welded in production but shall not be less than 3/16 inch.

Note 7. - Welders qualified under the requirements of test number 3 shall be considered qualified to make welds governed by test number 1 for the positions tested under test number 3.

Note 8. - Welders qualified under the requirements of test number 4 shall be considered qualified to make welds governed by tests numbers 1, 2, and 3 for the positions tested under test number 4.

Note 9. - Where the welder is to be qualified for welding or torch brazing of piping onboard ship the space restrictions shown in connection with tests numbers 3, 4, and 6 shall be used.

3.3 Test assemblies. -

3.3.1 Welder's identification on test assemblies. - Qualification test assemblies and all specimens removed therefrom shall be stamped, stenciled, or otherwise indelibly marked with the welder's identification number or symbol, and other suitable markings to sufficiently establish the position of welding and type of specimen.

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3.3.2 Requirements of test numbers 1, 2, 3 and 4.- Specimens cut from these tests shall be bent in a bending jig having the exact profile illustrated by figure 6. Specimens for qualification test number 1 modified for thinner material shall be bent in a jig having a contour proportional to the thickness of the test plates involved as shown in figure 6. Root bend specimens shall be bent with the root of the weld in tension; face bend specimens with the face in tension, and side bend specimens with that side, which is suspected of being the worst of the two, in tension.

3.3.2.1 Specimens shall be bent until they conform to the contour of the female member of the jig, and until a 1/32 inch wire cannot be placed between the specimen and any point on the curvature of the male member of the jig. Any specimen in which a crack or opening exists before the bending, or results from the bending, exceeding 1/8 inch measured in any direction shall be rejected. No elongation data are required.

3.3.2.2 In lieu of the procedure outlined above, welders may be qualified by radiographic examination of butt joints for qualification tests number 1, 2, 3 and 4. Satisfactory passing of tests shall be determined from radiograph which shows approved penetrameter and which shows deposited weld metal free from slag entrapment, weld or fusion zone cracking, and does not contain porosity in excess of that shown in applicable group of X-ray Standards for Production and Repair Welds (NAVSHIPS 250-692-2).

3.3.3 Requirements of test number 5.- Fillet weld specimens shall comply with the following requirements:

(a) Welds shall be of the specified size and shall be within the applicable convexity or concavity limits of the gage.
(b) Base metal shall be free from excessive undercutting.
(c) Surface of welds shall be free of cracks, porosity, overlap and roughness.
(d) Fractured surfaces of fillet welds shall be free from cracks, and the sum of lengths of slag inclusions, gas pockets, worm holes, and sections of incomplete fusion that are present shall not exceed 1-1/2 inches per 12 inches of weld.

3.3.4 Test number 6.- The torch brazed assembly shall comply with the following requirements:

(a) Exterior examination of the joint shall show a complete ring of filler metal between the tube and the outer ends of the fittings. Surfaces of the specimens shall be free of globules of filler metal. Fillets shall be concave and of minimum practicable dimension.
(b) The specimens shall be cut as shown in figure 8. Visual examination of the joints shall show complete penetration of the filler metal throughout at least 75 percent of the circumference of the fittings. Penetration is defined as follows: The presence or at least a metallic stain of filler metal at the fitting end of the tubing. The stain resulting from the penetration of the flux shall not be considered as penetration. The metallic stain is differentiated from the flux by its color which is grayish white. The flux stain is glossy in appearance.

3.3.5 Test number 7.- Specimens shall comply with the following requirements:

(a) Exterior examination of the joint shall show a complete line of filler metal at each edge of the joint. Surfaces of the specimen shall be free of globules of filler metal. Fillets shall be concave and of minimum practicable dimension.
(b) After visual examination specimens shall be fractured by peel test as shown in figure 10. Visual examination of the faying surface of the joint shall show at least 75 percent penetration of the filler metal throughout the joint. Specimens which do not fracture by peel test shall be acceptable if failure occurs in the base metal away from the joint.

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3.4 Filler metals. -

3.4.1 The following table shows the grouping of electrodes and welding rods together with applicable specifications. The number grouping of electrodes and welding rods shown in based essentially on their usability characteristics which fundamentally determine the ability of a welder to make satisfactory welds with a given electrode.

Table II - Grouping of electrodes and welding rods.

Type of weld deposit		Applicable specification	Electrode classification				Welding rod classification
			XX10 XX11	XX13	XX20 XX30	XX15 XX16	
Mild steel	All types	MIL-E-15599	M1	M1	M1		
Low alloy steel	MIL-180 MIL-230 MIL-260	MIL-E-986				M2	
Carbon molybdenum	All types	MIL-E-15716	M1		M1		
Chrome molybdenum	1/2 to 2 percent chrome	MIL-E-16589				M2	
Chrome molybdenum	2 to 10 percent chrome	MIL-E-16589				M2	
High alloy martensitic	12 chrome	MIL-E-16589				M2	
Chrome nickel	308, 309 312, 316 318, 347, 308 Mo. 308ELC 347LTA 349	MIL-E-16715				M3	
Chrome nickel	310 316ELC	MIL-E-16715				M3	
Mild steel	Class 1	MIL-R-908					M4

Note 1. - Qualification of a welder for M2, automatically qualifies him for M1 but not vice-versa. A welder qualifying with an electrode M3 and above is qualified only for that group.

Note 2. - Covering and filler metal types not listed above shall require separate qualification.

Note 3. - Base metal qualification of welding electrodes classes M1, M2 and M3 shall be performed on base metal of the comparable tensile strength and chemical analysis as the deposited weld where mechanical tests are used. Where qualification is by radiographic means carbon steel may be used as the base material for qualification tests with M1, M2 and M3 electrode classes.

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Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Copies of this standard for Military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C

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DATE OF BIRTH (Month) (Day) (Year) | PLACE OF BIRTH (City) (State) (Country) | CHECK NO.

WELDER'S EMPLOYMENT RECORD

PROCESS	FROM	TO

PHOTOGRAPH OR THUMB PRINT

DATE TAKEN _____

WELDER'S SYMBOL

CHECK:
RIGHT HAND ☐
LEFT HAND ☐

(Welder's Signature)

E-38261

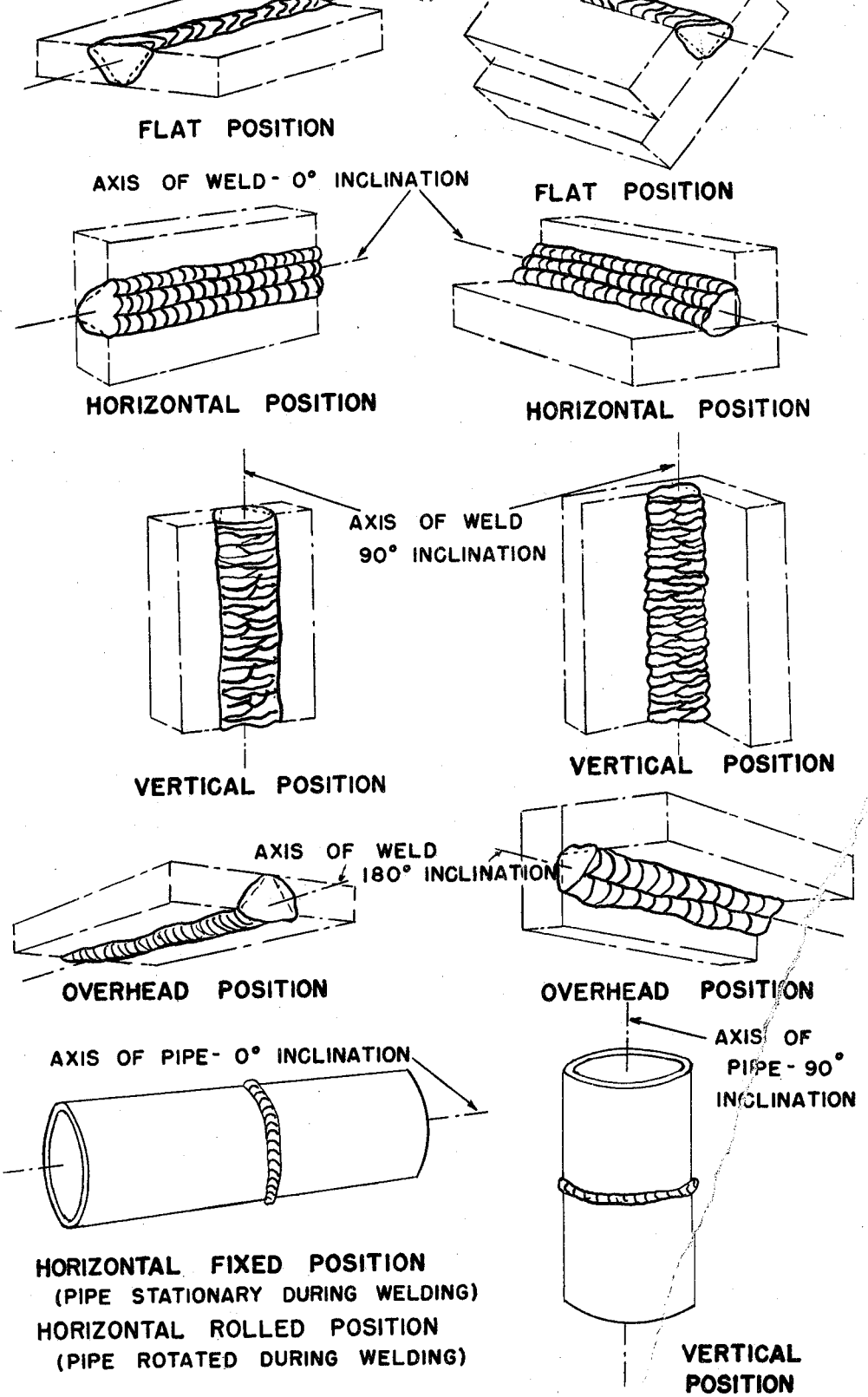
NAME (Last) (First) (Middle) | EMPLOYED BY | LOCATION OF PLANT

PROCESS	TEST NO.	BASE METAL	POSITION	FILLER METAL TYPE	FILLER METAL SIZE	DATE OF TEST	SIGNATURE OF INSPECTOR	AGENCY

RECORD CARD OF WELDER'S QUALIFICATION TESTS NAVSHIPS 4528 (1-54)

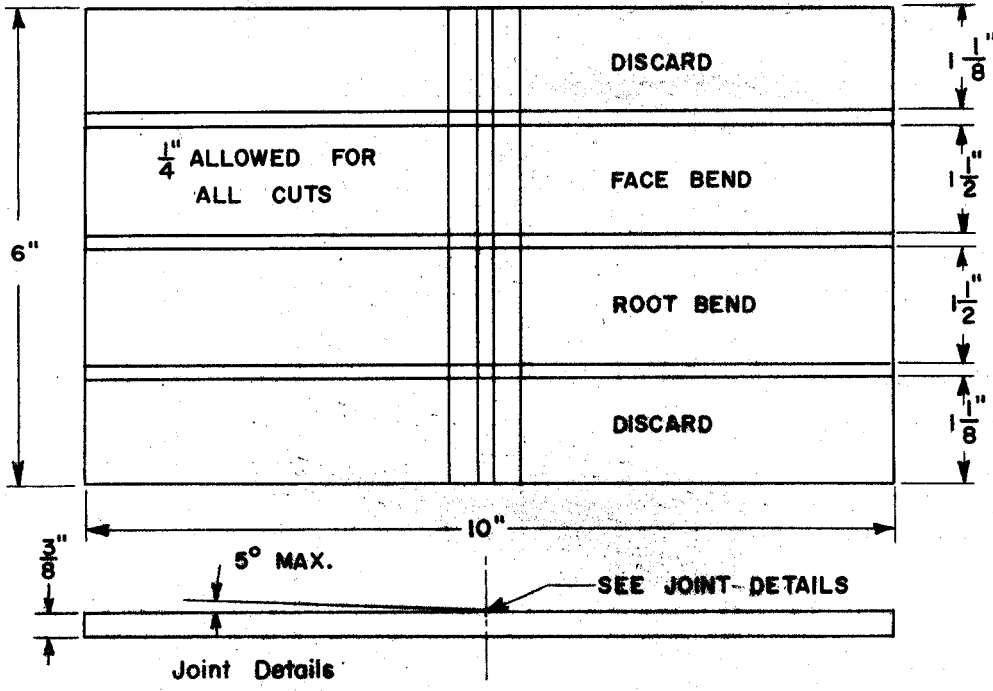
Figure 1. - Welder's identification and record of qualification tests.

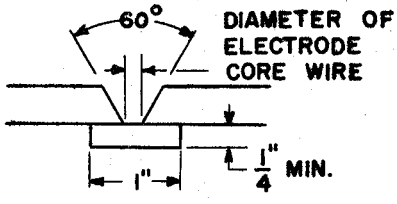
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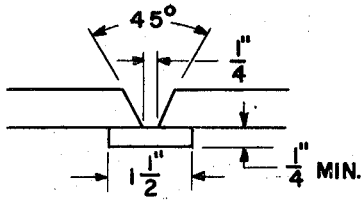
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Figure 2.-Positions of welding.

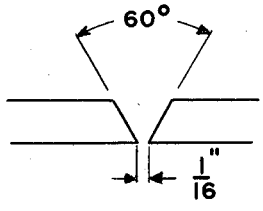




MANUAL SHIELDED METAL ARC



MANUAL AND SEMI-AUTOMATIC INERT GAS; METAL-ARC WELDING; MANUAL CARBON ARC WELDING; MANUAL GAS WELDING; SEMI-AUTOMATIC SUBMERGED ARC

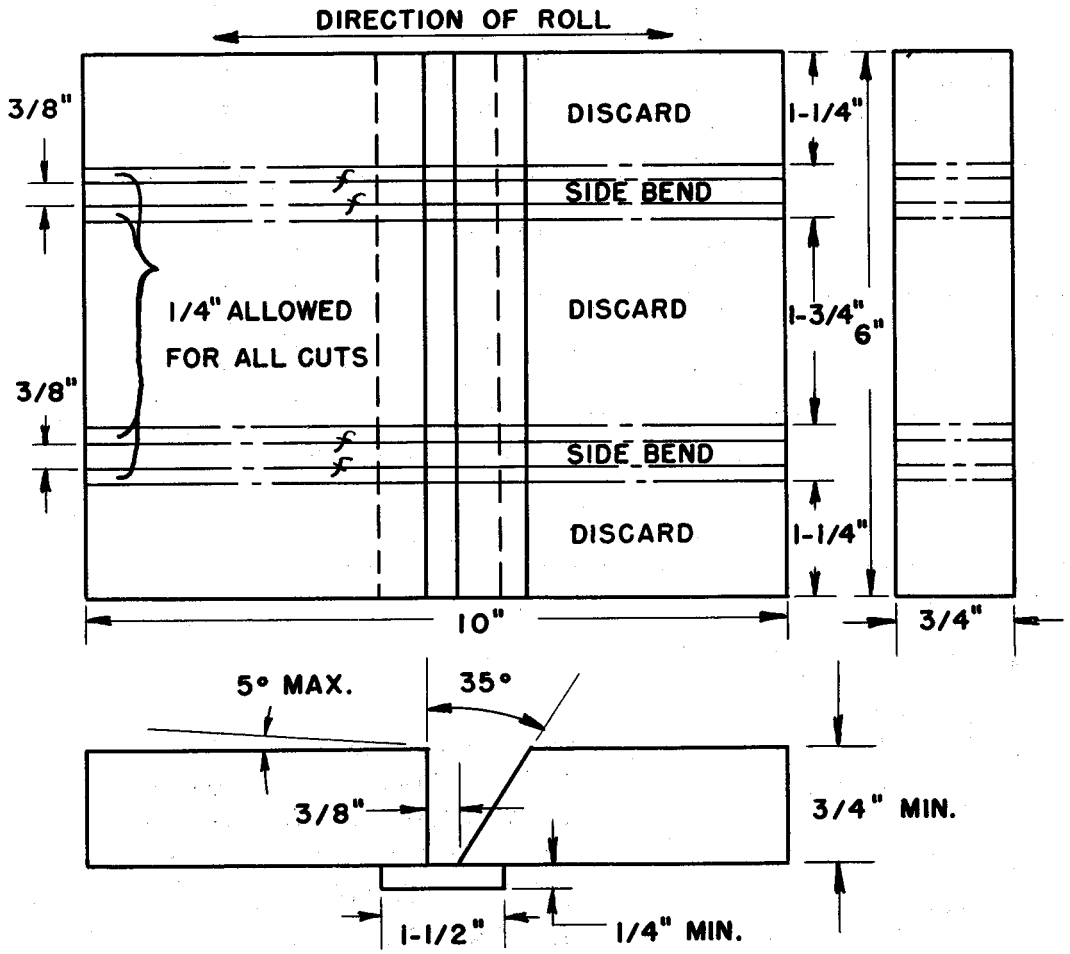


MANUAL GAS WELDING ALTERNATE

1. WELD SHALL BE MADE WITH THE MAXIMUM SIZE ELECTRODE OR WELDING ROD SUITABLE FOR THE POSITION AND THICKNESS OF TEST PLATE.
2. BACKING STRAP SHALL BE CONTIGUOUS WITH PLATES.
3. JOINTS WELDED IN VERTICAL POSITION SHALL BE WELDED UPWARDS.
4. WELDING SHALL BE DONE FROM ONE SIDE ONLY.
5. REINFORCEMENT AND/OR BACKING STRAP SHALL BE MACHINED FLUSH. UNDERCUTTING SHALL NOT BE REMOVED.
6. SPECIMENS MAY BE MACHINED OR OXYGEN CUT FROM PLATE.
7. EDGES OF SPECIMENS MAY BE BROKEN TO A MAXIMUM RADIUS OF T/6.

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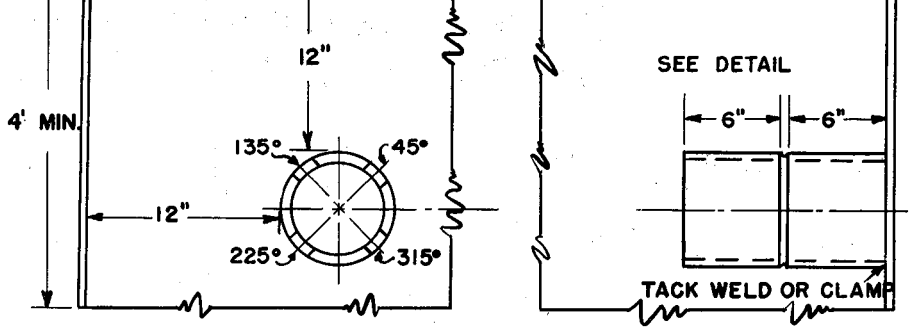
Figure 3.-Qualification test number 1



1. WELDS SHALL BE MADE WITH THE MAXIMUM SIZE ELECTRODE OR WELDING ROD SUITABLE FOR THE POSITION AND THICKNESS OF TEST PLATE.
2. BACKING STRAP SHALL BE CONTIGUOUS WITH PLATES.
3. JOINTS WELDED IN THE VERTICAL POSITION SHALL BE WELDED UPWARDS.
4. WELDING SHALL BE DONE FROM ONE SIDE ONLY.
5. MACHINE REINFORCEMENT AND BACKING STRAP FLUSH. DO NOT REMOVE ANY UNDERCUTTING.
6. SPECIMENS MAY BE MACHINED OR OXYGEN CUT FROM PLATE.
7. BREAK EDGES OF SPECIMENS TO A RADIUS OF T/6 MAXIMUM.

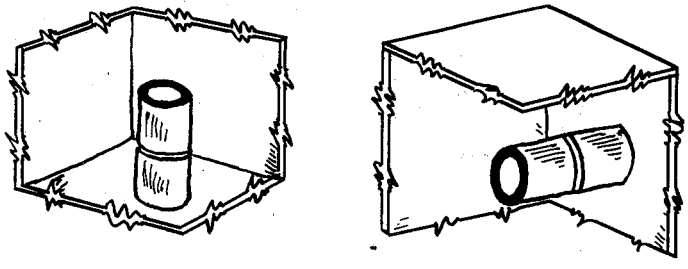
Figure 4.-Qualification test no. 2

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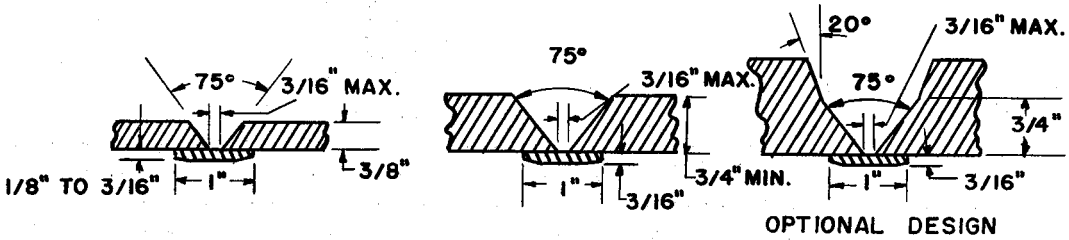


VERTICAL FIXED

HORIZONTAL FIXED



Joint Details



TEST NO. 3

FOR PIPING UP TO AND INCLUDING 3/4" THICK

OPTIONAL DESIGN

TEST NO. 4

FOR PIPING OVER 3/4" THICK

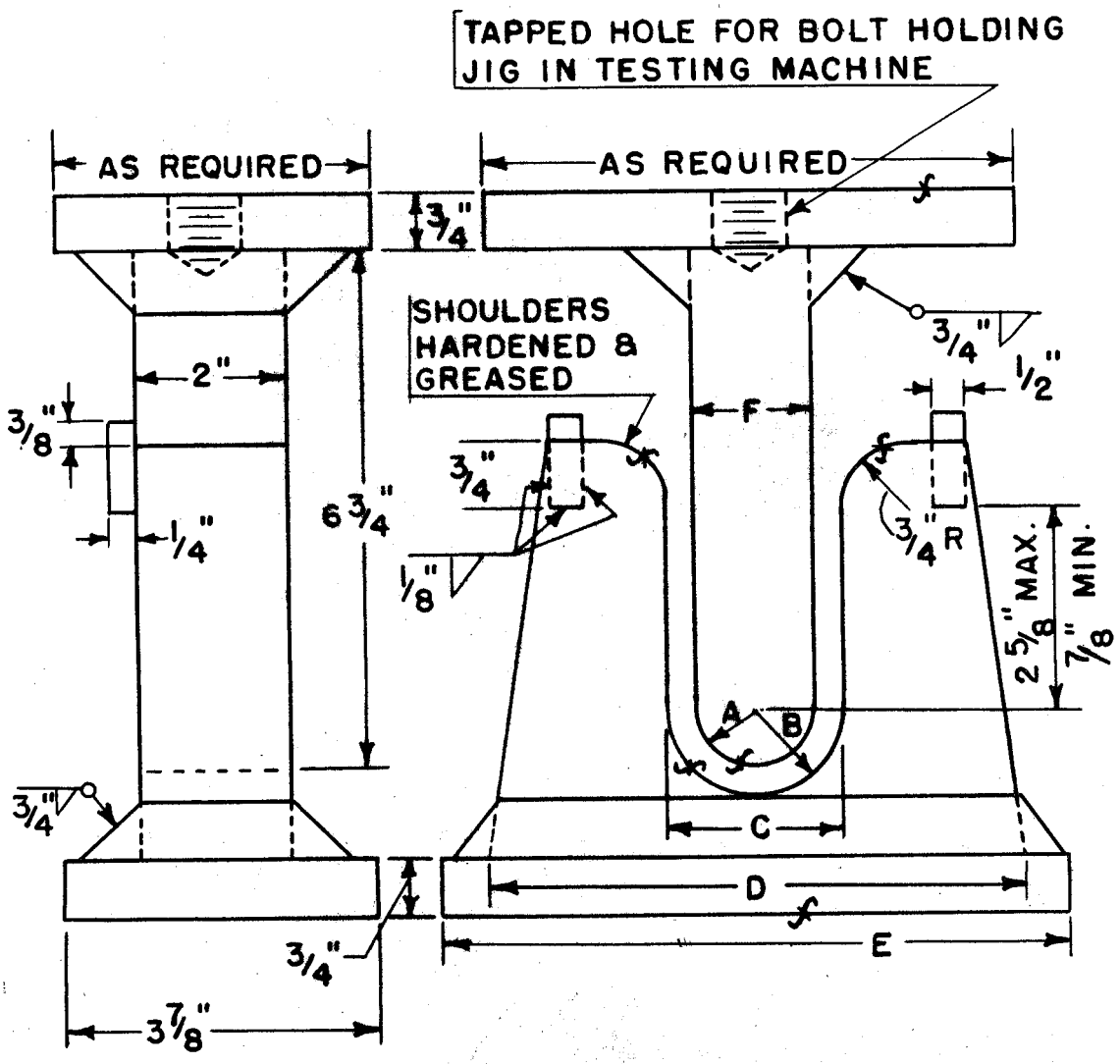
TEST 3A - WHERE MAXIMUM THICKNESS TO BE WELDED IS LESS THAN 3/8" WALL THICKNESS OF PIPE SHALL BE MAXIMUM TO BE WELDED BUT NOT LESS THAN 3/16".

SPECIMENS -
TEST 3 - FACE BENDS FROM 45° AND 225° POSITIONS
ROOT BENDS FROM 135° AND 315° POSITIONS
TEST 3A - MACRO SPECIMENS FROM 45°, 135°, 225°, AND 315° POSITIONS
TEST 4 - SIDE BENDS FROM 45°, 135°, 225°, AND 315° POSITIONS

1. SPACE RESTRICTIONS SHOWN MAY BE OMITTED WHEN WELDERS ARE TO WELD PIPING ONLY IN THE SHOP.
2. NOMINAL PIPE SIZE SHALL BE 5 INCHES OR LARGER.
3. WELD SHALL BE MADE WITH THE MAXIMUM SIZE ELECTRODE OR WELDING ROD SUITABLE FOR THE POSITION AND THICKNESS OF TEST PLATE.
4. WELDING SHALL BE FROM ONE SIDE ONLY.
5. MARK TOP AND FRONT OF PIPE TO INSURE LOCATION OF SPECIMENS.
6. MACHINE REINFORCEMENT AND BACKING RING FLUSH-UNDERCUTTING SHALL NOT BE REMOVED.
7. SPECIMENS MAY BE MACHINED OR OXYGEN CUT FROM PIPE.
8. EDGES OF SPECIMENS SHALL BE BROKEN TO A RADIUS OF T/6 MAXIMUM.

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Figure 5.- Qualification tests numbers 3 and 4.

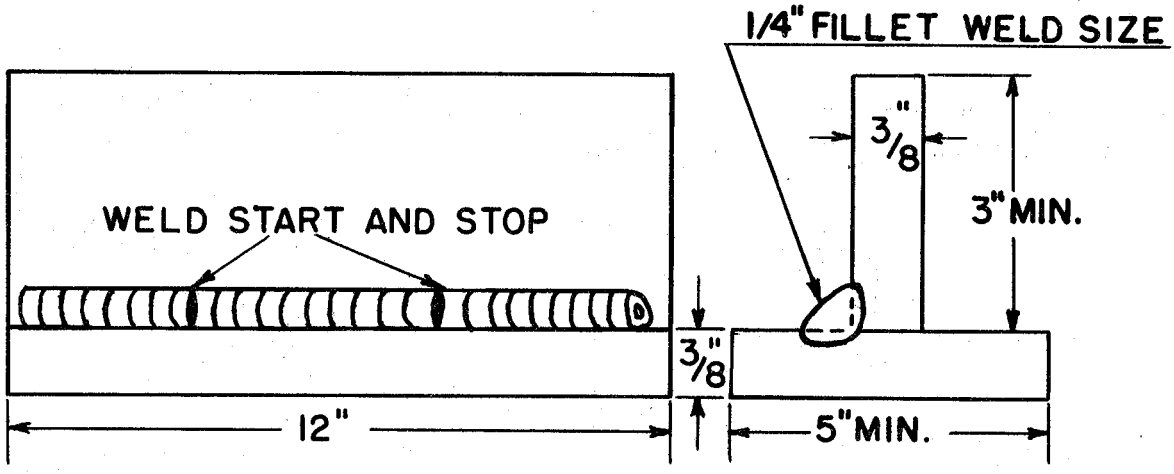


HARDENED ROLLS MAY BE USED ON SHOULDERS IF DESIRED

SPECIMEN MATERIAL	SPECIMEN THICKNESS	DIMENSIONS				
		A	B	C	D	E
BASE METALS OTHER THAN ALUMINUM	3/8"	3/4"	1-3/16"	2-3/8"	7-1/2	9
	T"	2T	3T+1/16	6T+1/8"	20T	24T
ALUMINUM	3/8"	1-1/8"	1-9/16"	3-1/8"	8-1/4"	9-3/4"
	T"	3T	4T+1/16	8T+1/8"	22T	26T

Figure 6.- Jig for guided bend test

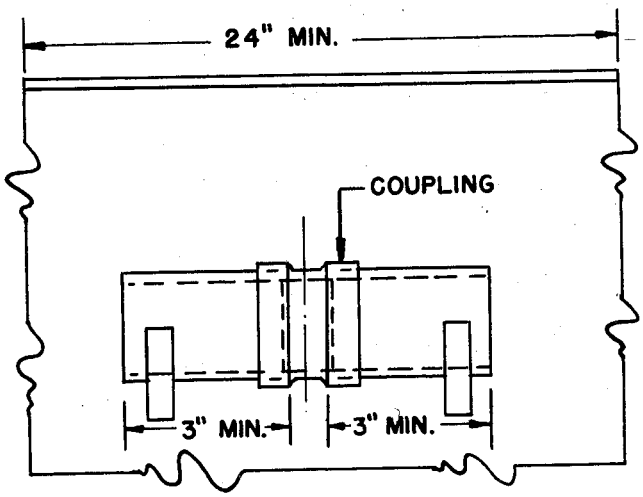
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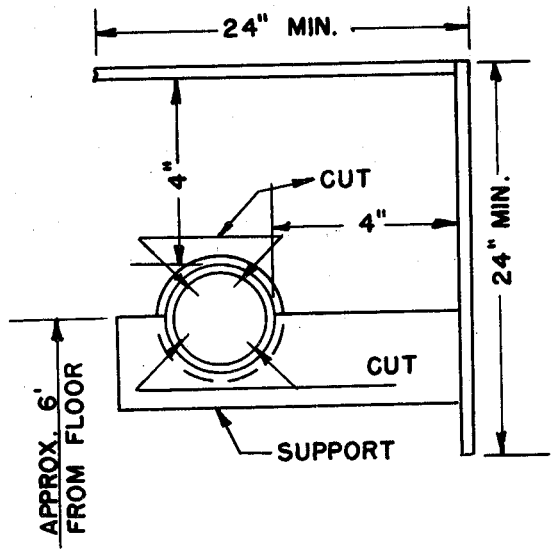
1. BASE METAL MAY BE CARBON STEEL.
2. WELD SHALL BE MADE WITH THE MAXIMUM SIZE OF ELECTRODE SUITABLE FOR THE POSITION AND THICKNESS OF TEST PLATE.
3. WELD SIZE SPECIFIED IS FOR SINGLE PASS WELD ONLY.
4. JOINTS WELDED IN THE VERTICAL POSITION OF WELDING SHALL BE WELDED UPWARDS.
5. BASE PLATE AND STANDING WEB SHALL BE STRAIGHT AND IN INTIMATE CONTACT THRUOUT ITS LENGTH AND SECURELY TACKED AT ENDS BEFORE FILLET WELD IS MADE.
6. THE TEST ASSEMBLY MAY BE MACHINED OR OXYGEN CUT INTO SECTIONS APPROXIMATELY 6 INCHES LONG TO FACILITATE TESTING.

Figure 7.-Qualification test number 5.

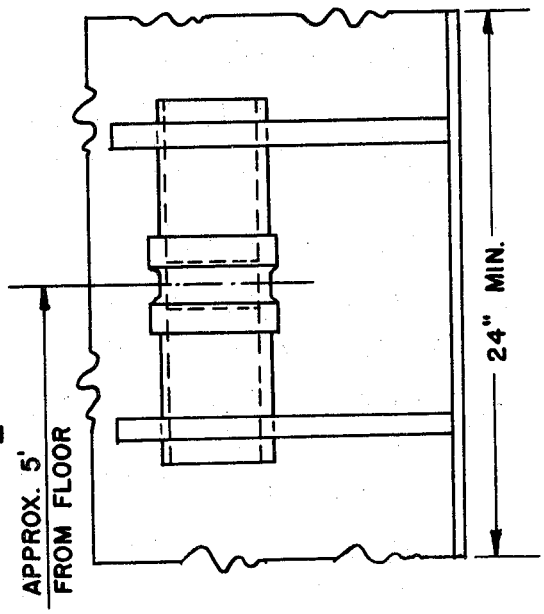
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HORIZONTAL POSITION



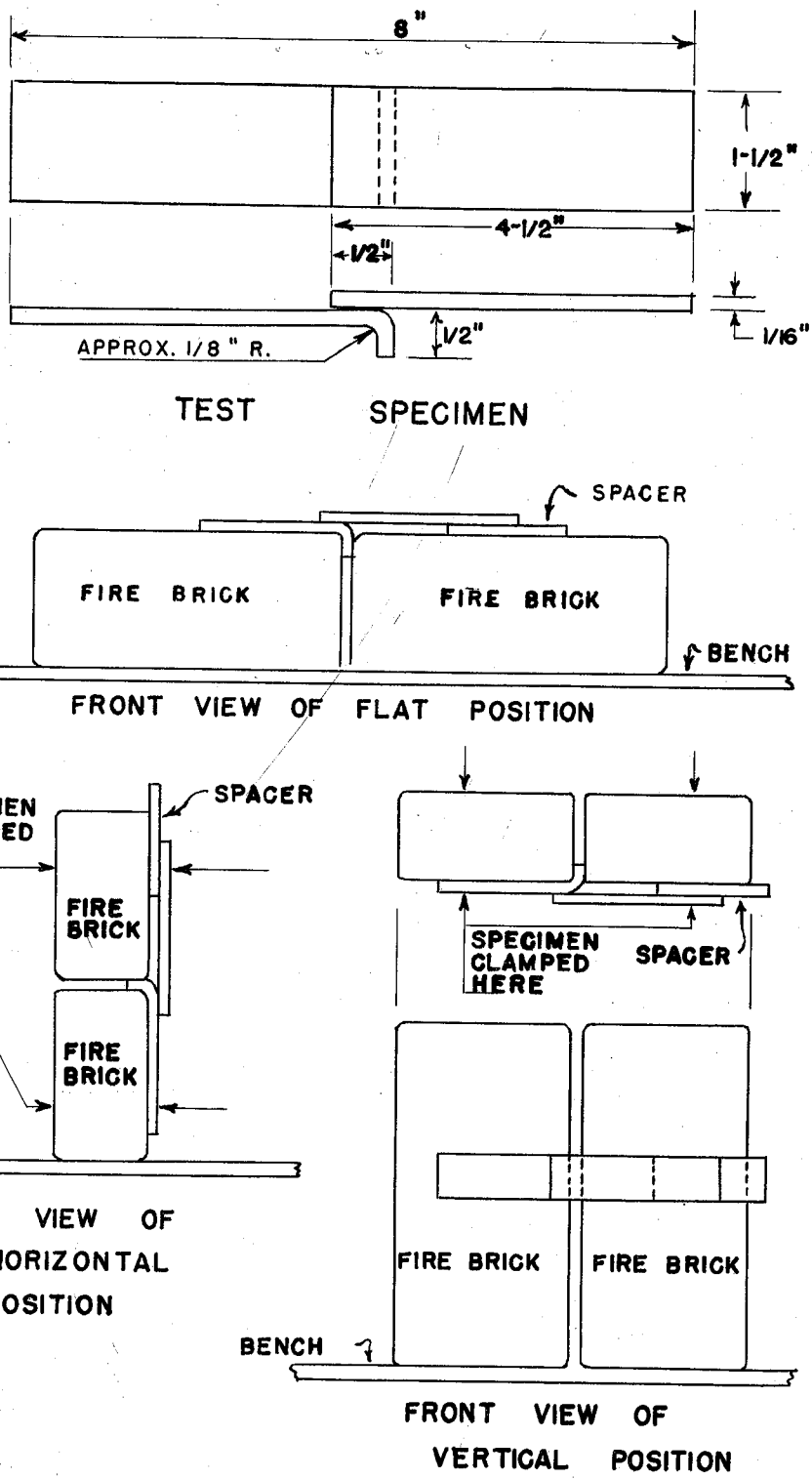
1. PIPE SIZES SHALL BE 2" MIN.
2. SPACE RESTRICTIONS SHOWN MAY BE OMITTED WHERE PIPING IS TO BE BRAZED ONLY IN THE SHOP.
3. FOR QUALIFICATION FOR BRAZING ON BOARD SHIP- BRAZE ONE COMPLETE COUPLING ASSEMBLY EACH IN OVERHEAD, HORIZONTAL AND VERTICAL POSITIONS. FOR QUALIFICATION FOR BRAZING IN SHOP- BRAZE ONE COUPLING ASSEMBLY WITH ONE JOINT IN HORIZONTAL AND ONE IN VERTICAL OVERHEAD POSITION.
4. SAW CUT PIPE AND COUPLING AT POINTS INDICATED FOR VISUAL AND MACRO EXAMINATION OF BRAZED JOINT.



VERTICAL POSITION

Figure 8.—Qualification test number 6.

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1. ASSEMBLY SHALL BE BRAZED IN HORIZONTAL, VERTICAL, OR FLAT POSITION AS APPLICABLE. HEATING SHALL BE DONE FROM ONE SIDE ONLY.
2. ASSEMBLY SHALL BE TESTED AS INDICATED IN FIG. 10

Figure 9.–Qualification test number 7.

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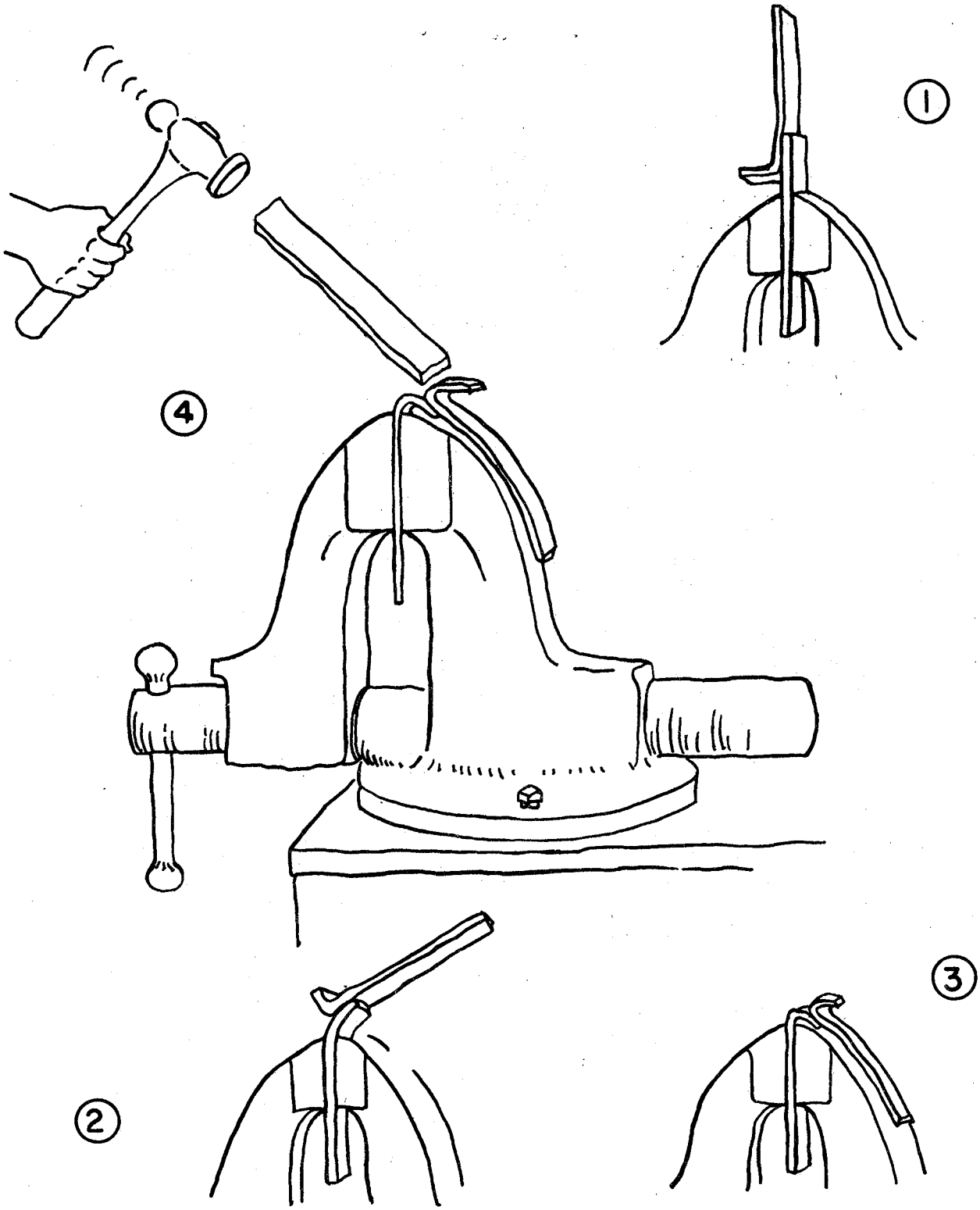


Figure 10.– Suggested method of testing specimens for qualification test number 7.

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